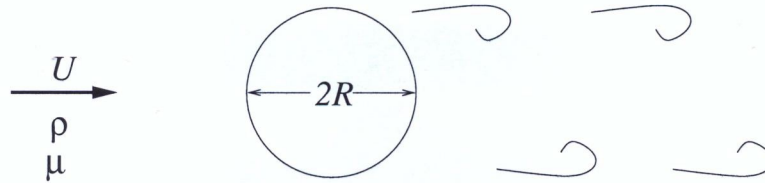


A wind stream of velocity U flows over a circular wire of radius R . It is observed that, when the velocity increases above a critical value U_c , the steady flow becomes unstable and it is replaced by a periodic solution characterized by the appearance of vortices of opposite circulation that come off the bottom and top of the cylinder in an alternative fashion, as shown in the figure below.

1. Use the dimensional analysis to reduce the parametric dependence of U_c .



The periodic flow that appears for $U > U_c$ causes a periodic force on the cylinder perpendicular to the incident wind stream.

2. Use again the Π theorem to simplify the parametric dependence of the period of the motion T that emerges for $U > U_c$ and of the mean value of the force per unit length acting on the wire, given by its quadratic average value $\bar{F} = \sqrt{\int_0^T F^2(t) dt / T}$.
3. Explain how the resulting expressions are simplified for large values of U such that $U \gg \nu/R$, where ν is the kinematic viscosity of air.

Experiments in a wind tunnel with a wire of radius $R = 1$ mm give $U_c = 0.2$ m/s along with the following values of T and $\bar{F}$ for $U > U_c$:

U (m/s)	T (ms)	$\bar{F}$ (mN/m)
0.2	100	0
0.5	25	0.01
1	10	0.12
2	5	0.48
4	2.5	1.92
10	1	12
20	0.5	48

Use this experimental information to determine, for a wire of radius $R = 2$ mm,

4. The value of U_c .
5. The variation of T and $\bar{F}$ with U for $U > U_c$.
6. Obtain the simplified expressions for T and $\bar{F}$ that apply for sufficiently large values of U .

① $U_c = f(\rho, \mu, R) \rightarrow \frac{U_c}{\mu/(8R)} = C_1^{\text{CONSTANT}}$

② $T = T(\rho, U, \mu, R) \rightarrow \frac{T}{R/U} = f_T\left(\frac{8UR}{\mu}\right)$
 $\bar{F} = \bar{F}(\rho, U, \mu, R) \rightarrow \frac{\bar{F}}{8U^2R} = f_F\left(\frac{8UR}{\mu}\right)$

③ IF $\frac{8UR}{\mu} \gg 1 \rightarrow \frac{T}{R/U} = C_T^{\text{CONSTANTS}}$
 $\frac{\bar{F}}{8U^2R} = C_F^{\text{CONSTANTS}}$

④ $\frac{U_{c1}}{\mu/R_1} = \frac{U_{c2}}{\mu/R_2} \rightarrow U_c = \frac{1}{2} \cdot 0.2 \text{ m/s} = 0.1 \text{ m/s}$

⑤ TO OBTAIN PHYSICAL SIMILARITY FOR THE CONDITIONS AT EACH EXPERIMENTAL POINT

$8 \frac{U_1 R_1}{\mu} = 8 \frac{U_2 R_2}{\mu} \rightarrow U_2 = \frac{1}{2} U_1$

$\frac{T_1}{R_1/U_1} = \frac{T_2}{R_2/U_2} \rightarrow T_2 = 4T_1$

$\frac{\bar{F}_1}{8U_1^2 R_1} = \frac{\bar{F}_2}{8U_2^2 R_2} \rightarrow \bar{F}_2 = \frac{\bar{F}_1}{2}$

$U(\frac{m}{s})$	$T(\text{ms})$	$\bar{F}(\text{mN/m})$
0.1	400	0
0.25	100	0.005
0.5	40	0.06
1	20	0.24
2	10	0.96
5	4	6
10	2	24

⑥ $\frac{8UR}{\mu} \gg 1$

$\frac{T}{R/U} = C_T = 10$

$\frac{\bar{F}}{8U^2R} = C_F = 0.1$

OBTAIND FROM EVALUATIONS OF THE EXPERIMENTAL DATA